Software Requirements Specification for Gold Star

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1 Introduction

1.1 Purpose

This document outlines the technical features and specifications of our web-based productivity and habit-building platform, *Gold Star*.

1.2 Scope

For those struggling with ADHD, building and maintaining habits and consistent routines can prove incredibly difficult. Changes or adaptations in routine become disruptive and often lead to abandoning the routine entirely. Gold Star is designed to mitigate these challenges by providing a platform to help users organize their schedules, break down tasks, and build sustainable routines (Faye et al., 2026).

1.3 Definitions, Acronyms, and Abbreviations

Activity Alternation - Switching from an activity requiring high mental focus to an activity requiring low mental focus

Activity Switching - The cognitive process of disengaging from one task and shifting mental focus to another task

Adaptive Scheduling - To schedule a task which can be completed anytime within a set timeframe. For example, a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

Application Programming Interface (API) - A set of standard rules, protocols, and definitions allowing different software applications to interact with one another

Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions

Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website

Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code

GitHub - A cloud-based platform where developers can collaborate on software code

Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD

Habit Stacking – Interspersing new habits with existing habits to make task completion easier without feeling overwhelmed

Hyperactivity - A state of being unusually or abnormally active

HyperText Markup Language (HTML) - The standard markup language used to create and structure webpages

JavaScript (JS) - A lightweight programming language used to build dynamic interactivity on web pages

Jest - An open-source JavaScript testing framework

JSDoc - An API documentation generator and syntax standard for JavaScript

Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats

PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data

Reinforce - To make something stronger, more effective, or more compelling by adding support

Relational Database Management System (RBMS) - Software used to store, manage, query, and retrieve data structured in table form

Rigid Scheduling – To schedule a high-priority task which must either be initiated at or completed before an established time. For example, a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.

Routine Adaption – A requirement for an individual to change one's schedule or create new routines

Routine - A sequence of actions regularly followed, or a fixed program

Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all

Visual Studio Code (VSCode) - A development environment used for software engineering and web development

(Faye et al., 2026)

1.4 References

Faye, E., Allen, M., Aaron, B., Bragg, M., Bucher, J., Carey, A., Looney, M., & Nebuchadnezzar, L. (2026). Lab 1 - gold star product description. https://mbrag003.github.io/CS410-411W-Gold/labs/

1.5 Overview

Section 2 gives an overview of Gold Star's purpose, function, and user roles.

2 Overall Description

2.1 Product Perspective

Gold Star is a web-based productivity and habit-building platform designed around individuals affected by ADHD symptoms. It creates an environment for individuals to create a schedule that works with them rather than against them, forming routines and invoking consistency. Gold star aims to introduce structure and provide a framework to overcome ADHD-related struggles like task initiation, consistency, commitment, and completion, ultimately resulting in improvements in productivity and a reduction of scheduling-related stress.

2.2 Product Functions

Gold Star's scheduling system provides a calendar view to allow users to get an overview of their upcoming or uncompleted tasks and a broader picture of their working schedule as a whole. Gold Star also implements an Adaptive Scheduling algorithm that works with the user to organize tasks and create a schedule based on their productivity level and the task's assigned priority. Task priority allows a user to determine the importance and urgency of a task. Through Habit Stacking, recognized habits are connected to pre-established routines, allowing consistently scheduled tasks to become apart of the Habit Stack rather than automatically becoming separate tasks.

2.3 User Characteristics

Gold Star has only one primary user role: ADHD-affected individuals looking to establish structure and organization into their life. Users manage tasks, including assigning them priorities, and make use of Gold Star to create a schedule accommodating of their productivity patterns. The scheduling system works with Adaptive Scheduling, Task Prioritization, and Habit Stacking to allow the flexibility needed for habit building. This is especially important in individuals with ADHD, so they aren't forced into a rigid routine that falls apart once an interruption occurs.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A